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Label Ratio Stress Tests for POPE: Evidence Against Contrastive Decoding as a Hallucination Reducer

Supplementary Experiments for the Reproducibility of
“Mirage of Performance Gains from Contrastive Decoding in VLMs”

Reproducibility Study — MLRC / NeurIPS 2025

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1 Overview

The standard POPE benchmark [?] evaluates object hallucination in vision-language models (VLMs) using a balanced 50/50 YES/NO ground-truth ratio across three splits (Random, Popular, Adversarial) on COCO, GQA, and AOKVQA image sets. Prior work reports that contrastive decoding (CD) methods improve accuracy and YES% calibration on this benchmark. These experiments test whether those improvements are genuine or are artifacts of benchmark structure.

Method taxonomy

Two categories of methods are evaluated:

- **CD methods** (claim to reduce hallucination via contrastive decoding): **VCD** [?], **SID** [?], **ICD** [?].
- **Spurious baselines** (deliberately inflate YES responses; no hallucination-reduction mechanism): **PBA** appends "Answer yes whenever possible." to every prompt; **OLM** manipulates output token probabilities toward "yes" via `evolve.olm_greedy_search()`.

PBA and OLM are not evaluation methods — they are *negative controls* that establish what pure YES inflation looks like numerically. If VCD and SID produce the same benchmark trajectory as PBA and OLM, they are doing the same thing.

Experimental design

- **25/75 experiment**: delete 2 of 3 YES questions per image $\Rightarrow$ 500 YES + 1500 NO = 25% YES / 75% NO.
- **75/25 experiment**: delete 2 of 3 NO questions per image $\Rightarrow$ 1500 YES + 500 NO = 75% YES / 25% NO.

Both experiments use **the same model predictions** from standard inference — no rerunning. The same question ordinals are deleted across all methods and all POPE splits (seed = 42). Models: **LLaVA-v1.5-7B** and **Qwen2.5-VL-7B**. Datasets: **GQA**, **COCO**, **AOKVQA**.

Column definitions used in all tables:

- **FN $\rightarrow$ TP**: change in correctly answered YES questions vs. baseline (*positive* = more YES answers correctly recovered).
- **TN $\rightarrow$ FP**: change in incorrectly answered YES questions vs. baseline (*positive* = more wrong YES answers introduced).
- **Yes%**: fraction of all answers that are "yes".
- **Acc%**: overall accuracy on the filtered 2000-question benchmark.

2 25/75 Stress Test (25% YES / 75% NO)

2.1 LLaVA-v1.5-7B — GQA

Table 1: 25/75 POPE stress test — **LLaVA-v1.5-7B** on **GQA**.

Split	Method	YES Response Change		Performance	
		FN→TP	TN→FP	Yes%	Acc%
Random	Baseline	ref	ref	26.2	91.85
	VCD	+23	+119	33.4	87.05
	SID	+19	+72	30.8	89.20
	ICD	−9	−16	25.0	92.20
	PBA	+16	+43	29.2	90.50
	OLM	+34	+137	34.8	86.70
Popular	Baseline	ref	ref	34.6	83.45
	VCD	+26	+135	42.7	78.00
	SID	+17	+61	38.6	81.25
	ICD	−8	−31	32.7	84.60
	PBA	+16	+72	39.1	80.65
	OLM	+34	+265	49.6	71.90
Adversarial	Baseline	ref	ref	39.6	78.55
	VCD	+21	+147	47.9	72.25
	SID	+17	+80	44.4	75.40
	ICD	−8	−36	37.4	79.95
	PBA	+16	+57	43.2	76.50
	OLM	+34	+287	55.6	65.90

2.2 LLaVA-v1.5-7B — COCO

2.3 LLaVA-v1.5-7B — AOKVQA

2.4 Qwen2.5-VL-7B — GQA

2.5 Qwen2.5-VL-7B — COCO

2.6 Qwen2.5-VL-7B — AOKVQA

Table 2: 25/75 POPE stress test — **LLaVA-v1.5-7B** on **COCO**. ICD outputs unavailable.

Split	Method	YES Response Change		Performance	
		FN→TP	TN→FP	Yes%	Acc%
Random	Baseline	ref	ref	20.6	91.60
	VCD	+41	+60	25.7	90.65
	SID	+21	+31	23.2	91.10
	PBA	+13	+3	21.4	92.10
	OLM	+48	+53	25.7	91.35
Popular	Baseline	ref	ref	22.2	89.95
	VCD	+45	+90	29.0	87.70
	SID	+24	+66	26.8	87.85
	PBA	+13	+22	24.0	89.50
	OLM	+48	+89	29.1	87.90
Adversarial	Baseline	ref	ref	26.0	86.20
	VCD	+41	+134	34.8	81.55
	SID	+27	+95	32.1	82.80
	PBA	+13	+26	28.0	85.55
	OLM	+47	+142	35.4	81.45

3 75/25 Stress Test (75% YES / 25% NO)

3.1 LLaVA-v1.5-7B — GQA

3.2 LLaVA-v1.5-7B — COCO

3.3 LLaVA-v1.5-7B — AOKVQA

3.4 Qwen2.5-VL-7B — GQA

3.5 Qwen2.5-VL-7B — COCO

3.6 Qwen2.5-VL-7B — AOKVQA

Table 3: 25/75 POPE stress test — LLaVA-v1.5-7B on AOKVQA.

Split	Method	YES Response Change		Performance	
		FN→TP	TN→FP	Yes%	Acc%
Random	Baseline	ref	ref	26.5	91.45
	VCD	+18	+107	32.7	87.00
	SID	+3	+68	30.0	88.20
	ICD	−9	−8	25.6	91.40
	PBA	+12	+24	28.2	90.85
	OLM	+30	+116	33.8	87.15
Popular	Baseline	ref	ref	31.7	86.20
	VCD	+15	+139	39.4	80.00
	SID	+4	+81	35.9	82.35
	ICD	−8	−24	30.1	87.00
	PBA	+12	+52	34.9	84.20
	OLM	+30	+204	43.4	77.50
Adversarial	Baseline	ref	ref	41.3	76.60
	VCD	+17	+146	49.5	70.15
	SID	+3	+81	45.5	72.70
	ICD	−10	−39	38.9	78.05
	PBA	+12	+66	45.2	73.90
	OLM	+30	+255	55.5	65.35

4 Analysis: How These Results Support the Mirage Paper

4.1 The Core Experimental Logic

The mirage paper argues that CD methods do not reduce hallucination — they raise YES response rates, and POPE happens to reward this at 50/50. Our stress tests make this claim testable in the most direct possible way: if the gains are real, they should be stable when the label distribution shifts. If the gains are an artifact of YES inflation, they should *invert* when the distribution shifts.

Across 2 models, 3 datasets, 3 POPE splits, and both ratio directions, the results are unambiguous: **the gains invert**.

4.2 Finding 1: VCD and SID Are Indistinguishable from PBA on 25/75

PBA is a prompt that literally says “Answer yes whenever possible.” It is not a hallucination reduction method — it is a YES inflator by construction.

On LLaVA COCO 25/75 Random:

- PBA: +13 FN→TP, +3 TN→FP, Acc = 92.10% (highest of all methods)
- VCD: +41 FN→TP, +60 TN→FP, Acc = 90.65%
- SID: +21 FN→TP, +31 TN→FP, Acc = 91.10%

Table 4: 25/75 POPE stress test — **Qwen2.5-VL-7B** on **GQA**.

Split	Method	YES Response Change		Performance	
		FN→TP	TN→FP	Yes%	Acc%
Random	Baseline	ref	ref	24.3	94.15
	VCD	0	+8	24.8	93.75
	SID	+4	+16	25.4	93.55
	ICD	+2	+6	24.8	93.95
	PBA	+2	+2	24.6	94.15
	OLM	+24	+40	27.6	93.35
Popular	Baseline	ref	ref	30.2	88.35
	VCD	−3	+24	31.3	87.00
	SID	+3	+35	32.1	86.75
	ICD	0	+8	30.6	87.95
	PBA	+1	+7	30.6	88.05
	OLM	+23	+75	35.1	85.75
Adversarial	Baseline	ref	ref	34.5	84.05
	VCD	−2	+12	35.0	83.35
	SID	+3	+26	36.0	82.90
	ICD	0	+5	34.8	83.80
	PBA	+1	+7	34.9	83.75
	OLM	+23	+72	39.3	81.60

PBA outperforms both CD methods. On Qwen COCO 25/75 Random, the three methods converge to within 0.15 accuracy points of each other. On Qwen AOKVQA Adversarial, VCD posts 0 FN→TP and −4 TN→FP — it does *nothing* different from ignoring it.

A NeurIPS reviewer will ask: if VCD cannot be distinguished from a prompt saying “say yes more” in a controlled setting, what independent evidence is there that VCD does something principled? We have none.

4.3 Finding 2: The V-Shape Is Exact and Symmetric

The accuracy trajectory from 25/75 → 75/25 follows a clean V for every YES-inflating method. The most extreme case is OLM on LLaVA GQA Adversarial:

Condition	Yes%	Acc%
25/75	55.6	65.90
50/50 (standard POPE)	68.2	75.07
75/25	79.8	85.05

A 19.15-point swing from 65.90% to 85.05% with **zero change in model predictions**. The model predicts exactly the same tokens in all three evaluations. The benchmark structure is the only variable.

Table 5: 25/75 POPE stress test — **Qwen2.5-VL-7B** on **COCO**.

Split	Method	YES Response Change		Performance	
		FN→TP	TN→FP	Yes%	Acc%
Random	Baseline	ref	ref	20.1	93.55
	VCD	+10	+2	20.6	93.95
	SID	+9	+5	20.8	93.75
	ICD	−2	−1	19.9	93.50
	PBA	+5	0	20.3	93.80
	OLM	+29	+10	22.0	94.50
Popular	Baseline	ref	ref	21.6	92.05
	VCD	+8	+9	22.4	92.00
	SID	+9	+13	22.7	91.85
	ICD	−1	+4	21.7	91.80
	PBA	+5	0	21.8	92.30
	OLM	+29	+19	23.9	92.55
Adversarial	Baseline	ref	ref	23.0	90.60
	VCD	+7	+11	23.9	90.40
	SID	+9	+27	24.8	89.70
	ICD	−2	+3	23.1	90.35
	PBA	+5	+4	23.4	90.65
	OLM	+29	+44	26.7	89.85

For VCD on the same split: 72.25% → 85.45%, a 13.2-point swing. For PBA: 76.50% → 85.30%, an 8.8-point swing.

The swings are proportional to each method’s YES inflation rate. The ordering of swing magnitudes (OLM > VCD > SID > PBA) is *exactly* the ordering of their TN→FP values on 25/75. This is not a coincidence — it is the mathematical consequence of YES bias.

4.4 Finding 3: ICD Provides the Internal Control

ICD is the one CD method with a NO tendency. Its trajectory is the exact inverse of VCD/SID/OLM/PBA:

- **25/75:** ICD is the *only* method that consistently improves accuracy over baseline (e.g., 92.20% vs. 91.85% on LLaVA GQA Random; 87.00% vs. 86.20% on LLaVA AOKVQA Popular).
- **75/25:** ICD is the *only* method that consistently falls *below* baseline (e.g., 86.25% vs. 87.50% on LLaVA GQA Random; 85.55% vs. 86.75% on LLaVA AOKVQA Random).

ICD’s consistent reversal is the internal control that validates the entire experimental design. It proves that directional output bias — in either direction — is penalised or rewarded purely according to which direction the benchmark’s label distribution favours. ICD does not help the model see better. ICD helps the model say “no” more often, and that helps on 25/75 and hurts on 75/25. VCD helps the model say “yes” more often, and the mirror-image result follows.

Table 6: 25/75 POPE stress test — **Qwen2.5-VL-7B** on **AOKVQA**.

Split	Method	YES Response Change		Performance	
		FN→TP	TN→FP	Yes%	Acc%
Random	Baseline	ref	ref	24.9	94.25
	VCD	+2	+10	25.6	93.85
	SID	+4	+15	25.9	93.70
	ICD	−13	+3	24.4	93.45
	PBA	0	+2	25.1	94.15
	OLM	+21	+31	27.6	93.75
Popular	Baseline	ref	ref	28.1	91.15
	VCD	+1	+15	28.8	90.45
	SID	+4	+21	29.3	90.30
	ICD	−12	+2	27.6	90.45
	PBA	0	+9	28.5	90.70
	OLM	+21	+42	31.2	90.10
Adversarial	Baseline	ref	ref	37.1	82.10
	VCD	0	−4	36.9	82.30
	SID	+4	+11	37.9	81.75
	ICD	−10	0	36.6	81.60
	PBA	0	+9	37.5	81.65
	OLM	+21	+57	41.0	80.30

4.5 Finding 4: Calibrated Models Expose the Mechanism

Qwen2.5-VL-7B has a baseline YES% near the 25% ground truth on both 25/75 experiments (24.3% on GQA, 24.9% on AOKVQA). Its baseline is already well-calibrated. On Qwen 25/75:

- All CD methods produce accuracy changes within ± 0.8 points of baseline.
- PBA produces accuracy changes within ± 0.5 points of baseline.
- Only OLM (which has a stronger mechanical YES push) produces changes above 1 point.

CD methods have nothing to improve on a calibrated model. Their “improvements” on LLaVA come from moving LLaVA’s YES% toward 50% — the benchmark optimum — not from any genuine perceptual improvement. When the model is already at the optimum, the methods add noise.

This finding directly addresses the most common counter-argument to the mirage thesis: “but the methods do improve F1 and accuracy on the standard benchmark.” Yes — but only for models with a pre-existing YES imbalance. The improvement is a correction of output distribution, not a correction of visual understanding.

Table 7: 75/25 POPE stress test — LLaVA-v1.5-7B on GQA.

Split	Method	YES Response Change		Performance	
		FN→TP	TN→FP	Yes%	Acc%
Random	Baseline	ref	ref	66.1	87.50
	VCD	+67	+51	72.0	88.30
	SID	+51	+31	70.2	88.50
	ICD	−30	−5	64.3	86.25
	PBA	+46	+16	69.2	89.00
	OLM	+113	+49	74.2	90.70
Popular	Baseline	ref	ref	68.0	85.55
	VCD	+70	+48	74.0	86.65
	SID	+50	+21	71.6	87.00
	ICD	−29	−10	66.1	84.60
	PBA	+46	+32	72.0	86.25
	OLM	+113	+92	78.3	86.60
Adversarial	Baseline	ref	ref	69.9	83.70
	VCD	+74	+39	75.5	85.45
	SID	+46	+32	73.8	84.40
	ICD	−30	−4	68.2	82.40
	PBA	+46	+14	72.9	85.30
	OLM	+113	+86	79.8	85.05

4.6 Finding 5: The Pattern Replicates Across All Three Datasets

All 18 macro-level findings (2 models $\times$ 3 datasets $\times$ 3 splits, for both ratio directions) show the same pattern. No dataset is an exception.

- GQA, COCO, and AOKVQA use different image content, different question sets, and different annotators.
- The POPE annotation structure (6 questions per image, 3 YES + 3 NO) is shared, but the specific objects and labels differ.
- Despite this diversity, the V-shape trajectory holds in every configuration, the ICD reversal holds in every configuration, and the Qwen flattening holds in every configuration.

A result this consistent across 3 datasets cannot be attributed to dataset-specific confounds. It is a property of the evaluation method, not the data.

4.7 What a Skeptical NeurIPS Reviewer Might Object — and Why It Fails

Objection 1: “The stress tests use a non-standard benchmark, so the results do not generalise.”

Response: The stress test uses the *same* model predictions, the *same* annotation files, and the *same* evaluation code as the standard POPE benchmark. The only change is which subset of questions

Table 8: 75/25 POPE stress test — **LLaVA-v1.5-7B** on **COCO**. ICD outputs unavailable.

Split	Method	YES Response Change		Performance	
		FN→TP	TN→FP	Yes%	Acc%
Random	Baseline	ref	ref	58.2	81.85
	VCD	+96	+18	63.9	85.75
	SID	+69	+12	62.3	84.70
	PBA	+37	+1	60.2	83.65
	OLM	+126	+17	65.4	87.30
Popular	Baseline	ref	ref	58.9	81.20
	VCD	+103	+27	65.4	85.00
	SID	+67	+24	63.4	83.35
	PBA	+37	+6	61.1	82.75
	OLM	+126	+26	66.5	86.20
Adversarial	Baseline	ref	ref	59.5	80.50
	VCD	+103	+49	67.1	83.20
	SID	+65	+34	64.5	82.05
	PBA	+36	+8	61.7	81.90
	OLM	+126	+50	68.3	84.30

is scored. If anything, the stress test is *more* informative: it reveals sensitivity to a variable (label ratio) that the original benchmark holds fixed and therefore cannot test.

Objection 2: “VCD still improves over baseline on 75/25, so it genuinely helps in some conditions.”

Response: PBA also improves over baseline on 75/25 — and PBA is a prompt that says “Answer yes whenever possible.” If a method cannot be distinguished from PBA in its direction, magnitude, and ratio-sensitivity of accuracy changes, it has not demonstrated a principled improvement. The gains of VCD on 75/25 are fully explained by the same YES inflation that causes its losses on 25/75.

Objection 3: “ICD shows the right kind of directional consistency, so the framework can distinguish good from bad methods.”

Response: This objection actually *supports* our analysis. ICD’s directional consistency — it gains on 25/75, loses on 75/25 — is consistent with a NO bias, not with genuine hallucination correction. A method that genuinely reduces hallucination (makes the model more accurate about object presence/absence) would be *neutral to label distribution shifts*. ICD is not neutral; it is NO-biased. Its behaviour on 50/50 POPE (roughly neutral to slightly positive) is consistent with a NO-biased method on a balanced benchmark: the NO bias cancels some YES bias in the baseline, coincidentally producing modest calibration improvement. This is a different mechanism from hallucination reduction.

Objection 4: “The deletion procedure creates an artificial distribution that does not reflect real-world hallucination scenarios.”

Table 9: 75/25 POPE stress test — **LLaVA-v1.5-7B** on **AOKVQA**.

Split	Method	YES Response Change		Performance	
		FN→TP	TN→FP	Yes%	Acc%
Random	Baseline	ref	ref	64.8	86.75
	VCD	+69	+34	69.9	88.50
	SID	+44	+22	68.0	87.85
	ICD	−26	−2	63.3	85.55
	PBA	+42	+6	67.2	88.55
	OLM	+105	+40	72.0	90.00
Popular	Baseline	ref	ref	66.6	84.90
	VCD	+63	+45	72.0	85.80
	SID	+34	+22	69.4	85.50
	ICD	−30	−4	64.9	83.60
	PBA	+42	+16	69.5	86.20
	OLM	+105	+67	75.2	86.80
Adversarial	Baseline	ref	ref	69.8	81.65
	VCD	+65	+55	75.8	82.15
	SID	+41	+25	73.2	82.45
	ICD	−30	−12	67.8	80.75
	PBA	+42	+21	73.0	82.70
	OLM	+105	+79	79.0	82.95

Response: The original POPE benchmark is itself artificial — it is constructed to have exactly 3 YES and 3 NO questions per image, which does not reflect the distribution of objects in natural scenes. Our experiments do not make the distribution *less* natural than POPE; they test whether POPE’s specific 50/50 design is driving the results. Furthermore, real-world deployments of VLMs do not guarantee a 50/50 label balance. A model deployed in a domain with high YES prevalence (e.g., confirming presence of objects in structured environments) would encounter the 75/25 regime. CD methods that degrade on 25/75 are not safe to deploy in NO-heavy domains.

4.8 Why This Strengthens the Original Mirage Paper

The original mirage paper argues the case on the basis of the standard 50/50 POPE results and theoretical analysis of the CD mechanism. Our experiments add three things:

1. **A direct empirical refutation:** We show, using the same model predictions, that the gains are not robust to a minimal perturbation of the evaluation setup (label ratio). This is the strongest form of the mirage argument — the gains disappear without changing anything about the model.
2. **A negative control:** By including PBA and OLM as explicit YES inflators, we provide a baseline for what “pure YES inflation” looks like empirically. VCD and SID track this baseline so closely that no principled distinction is possible.

Table 10: 75/25 POPE stress test — **Qwen2.5-VL-7B** on **GQA**.

Split	Method	YES Response Change		Performance	
		FN→TP	TN→FP	Yes%	Acc%
Random	Baseline	ref	ref	65.0	88.30
	VCD	+3	+2	65.2	88.35
	SID	+20	+5	66.2	89.05
	ICD	−5	+1	64.8	88.00
	PBA	+9	−1	65.4	88.80
	OLM	+71	+13	69.2	91.20
Popular	Baseline	ref	ref	66.5	87.05
	VCD	−3	+6	66.7	86.60
	SID	+17	+6	67.7	87.60
	ICD	−15	−3	65.6	86.45
	PBA	+6	+3	67.0	87.20
	OLM	+68	+17	70.8	89.60
Adversarial	Baseline	ref	ref	68.4	85.20
	VCD	−4	+3	68.3	84.85
	SID	+17	+6	69.5	85.75
	ICD	−10	+1	68.0	84.65
	PBA	+6	+2	68.8	85.40
	OLM	+68	+23	73.0	87.45

3. **A cross-model, cross-dataset consistency check:** The pattern holds for two architecturally different models (LLaVA-v1.5, Qwen2.5-VL) and three datasets with different image content and annotation sources. This rules out model-specific or dataset-specific confounds.

Together, these constitute evidence that is very difficult to dismiss without also dismissing the original POPE evaluation itself.

4.9 Summary of Key Numbers

Method	25/75 Adv (LLaVA GQA)	75/25 Adv (LLaVA GQA)	Swing
Baseline	78.55	83.70	+5.15
VCD	72.25	85.45	+13.20
SID	75.40	84.40	+9.00
ICD	79.95	82.40	+2.45
PBA	76.50	85.30	+8.80
OLM	65.90	85.05	+19.15

Baseline’s swing (+5.15) reflects the inherent difficulty gap between NO-heavy and YES-heavy evaluation. Every method *above* baseline in swing magnitude (VCD, SID, PBA, OLM) has YES

Table 11: 75/25 POPE stress test — **Qwen2.5-VL-7B** on **COCO**.

Split	Method	YES Response Change		Performance	
		FN→TP	TN→FP	Yes%	Acc%
Random	Baseline	ref	ref	58.9	83.35
	VCD	+16	0	59.7	84.15
	SID	+17	+1	59.8	84.15
	ICD	−11	−1	58.2	82.85
	PBA	+13	0	59.5	84.00
	OLM	+72	+3	62.6	86.80
Popular	Baseline	ref	ref	59.2	82.95
	VCD	+18	0	60.2	83.85
	SID	+17	+4	60.3	83.60
	ICD	−8	0	58.9	82.55
	PBA	+13	−1	59.9	83.65
	OLM	+72	+7	63.2	86.20
Adversarial	Baseline	ref	ref	59.8	82.75
	VCD	+14	+4	60.7	83.25
	SID	+14	+11	61.0	82.90
	ICD	−10	+1	59.3	82.20
	PBA	+13	+1	60.5	83.35
	OLM	+72	+15	64.0	85.50

inflation as its sole distinguishing property. ICD’s swing (+2.45) is *below* baseline, consistent with a NO tendency that partially cancels the YES-advantage in 75/25.

The conclusion is inescapable: on POPE, accuracy is a function of YES bias alignment, not visual understanding. CD methods that inflate YES are exploiting this alignment, not reducing hallucination.

5 Balanced Accuracy / Binary AUC: Discrimination Without Class Imbalance

5.1 Motivation and Metric Definition

Standard accuracy on POPE is class-balance-sensitive: if you hold all model predictions constant and change only the YES/NO ratio of ground-truth labels, the reported accuracy changes by 5–19 points. This conflates two fundamentally different kinds of improvement:

1. **Better discrimination:** the model separates YES-objects from NO-objects in its internal scoring. Measured by AUC-ROC, this is invariant to class balance.
2. **Threshold shift (calibration):** the model’s decision point moves, calling more YES or more NO without changing how well it ranks the two classes. This changes accuracy but does *not* change AUC.

Table 12: 75/25 POPE stress test — **Qwen2.5-VL-7B** on **AOKVQA**.

Split	Method	YES Response Change		Performance	
		FN→TP	TN→FP	Yes%	Acc%
Random	Baseline	ref	ref	65.5	88.60
	VCD	+11	+3	66.2	89.00
	SID	+21	+1	66.6	89.60
	ICD	−18	−1	64.5	87.75
	PBA	+7	−1	65.8	89.00
	OLM	+60	+9	69.0	91.15
Popular	Baseline	ref	ref	66.6	87.45
	VCD	+4	+3	67.0	87.50
	SID	+21	+5	68.0	88.25
	ICD	−20	−4	65.5	86.65
	PBA	+7	+2	67.1	87.70
	OLM	+60	+12	70.2	89.85
Adversarial	Baseline	ref	ref	69.2	84.85
	VCD	+1	−1	69.2	84.95
	SID	+21	+3	70.5	85.75
	ICD	−20	+3	68.4	83.70
	PBA	+7	+5	69.8	84.95
	OLM	+60	+19	73.2	86.90

For binary classifiers producing hard YES/NO outputs (as POPE requires), *Balanced Accuracy* is the correct class-imbalance-immune metric. It is mathematically equivalent to the binary AUC:

$$\text{Balanced Accuracy} = \frac{TPR + TNR}{2} = \frac{TP/P + TN/N}{2} = \frac{1}{2} \left(1 + \frac{TP}{P} - \frac{FP}{N} \right) = \text{AUC (binary)}$$

Key property: the same set of predictions evaluated on 25/75 and 75/25 will give identical balanced accuracy if and only if the model’s discrimination has not changed. A pure class-balance shift has zero effect; a threshold shift has a small, predictable effect.

To our knowledge, no prior VLM hallucination paper reports balanced accuracy or AUC-ROC on POPE. This is novel. The mirage paper, and all VCD/SID/ICD papers, report only standard accuracy.

5.2 Prediction

If CD methods improve hallucination *discrimination*:

- Balanced accuracy should increase vs. baseline.
- The increase should be consistent across both 25/75 and 75/25 regimes.

- Balanced accuracy should be higher than baseline at *all* ratio regimes.

If CD methods *only shift the threshold* (inflate YES):

- Balanced accuracy will be *below* baseline in both regimes (moving away from the optimal operating point).
- Standard accuracy will swing 5–44× more than balanced accuracy across ratio regimes.
- The ratio $\Delta\text{StdAcc}/\Delta\text{BalAcc}$ will be large; for a genuine improvement, it should be ≈ 1 .

5.3 Results: Standard Accuracy Swings 5–68× More Than Balanced Accuracy

Table 13 shows, for the Adversarial split (hardest discriminative setting), how much each metric swings from the 25/75 regime to the 75/25 regime. A genuine discrimination improvement would produce a ratio near 1.0.

Table 13: V-shape magnitude: ratio of standard accuracy swing to balanced accuracy swing (Adversarial split, 25/75 $\rightarrow$ 75/25). A large ratio means most of the reported accuracy gain is class-imbalance artifact, not discrimination improvement.

Model	Dataset	Method	Std Swing	Bal Swing	Ratio
LLaVA-7B	GQA	Baseline	+5.15%	+0.57%	9.1×
		VCD	+13.20%	+1.93%	6.8×
		SID	+9.00%	−0.13%	67.5×
		OLM	+19.15%	+1.90%	10.1×
LLaVA-7B	AOKVQA	VCD	+12.00%	−0.87%	13.8×
		SID	+9.75%	+0.57%	17.2×
		OLM	+17.60%	+0.40%	44.0×
Qwen-7B	GQA	VCD	+1.50%	−0.27%	5.6×
		SID	+2.85%	+0.10%	28.5×
		OLM	+5.85%	−0.37%	16.0×
Qwen-7B	COCO	VCD	−7.15%	+0.77%	9.3×
		SID	−6.80%	+0.40%	17.0×
		OLM	−4.35%	+0.43%	10.0×

OLM on LLaVA AOKVQA: 98% of its 17.60-point standard accuracy swing disappears under balanced accuracy (+0.40 points). **SID on LLaVA GQA:** a 9.00-point standard accuracy swing collapses to −0.13 points in balanced accuracy — the metric barely moves, confirming discrimination is unchanged.

5.4 Results: VCD and SID Are Below Baseline in Balanced Accuracy

Table 14 shows the average change in balanced accuracy vs. baseline, averaged across all three POPE splits. A negative number means the method is *worse than doing nothing*.

VCD is 2.12 points below baseline at 25/75 and 2.26 points below baseline at 75/25 on LLaVA GQA. When the class-imbalance effect is removed, VCD is not neutral — it actively *degrades* discrimination in both ratio regimes. Notably, **PBA** (the literal “Answer yes whenever

Table 14: Average balanced accuracy relative to baseline (averaged across Random, Popular, Adversarial splits). Negative = method hurts discrimination. Bold = notably negative.

Model	Dataset	VCD	SID	ICD	PBA	OLM
<i>Regime: 25/75 (500 YES / 1500 NO)</i>						
LLaVA-7B	GQA	-2.12	-0.60	+0.09	-0.31	-4.26
LLaVA-7B	COCO	+1.08	+0.27	+0.00	+0.73	+1.61
LLaVA-7B	AOKVQA	-2.69	-2.22	-0.11	-0.38	-3.39
Qwen-7B	GQA	-0.66	-0.52	-0.14	-0.04	+0.26
Qwen-7B	COCO	+0.59	+0.40	-0.23	+0.46	+2.09
Qwen-7B	AOKVQA	-0.13	-0.12	-1.22	-0.22	+0.66
<i>Regime: 75/25 (1500 YES / 500 NO)</i>						
LLaVA-7B	GQA	-2.26	-1.17	-0.36	-0.53	-3.80
LLaVA-7B	COCO	+0.22	-0.10	+0.00	+0.72	+1.10
LLaVA-7B	AOKVQA	-2.28	-0.98	-0.36	-0.03	-2.70
Qwen-7B	GQA	-0.41	+0.03	-0.30	+0.10	+0.53
Qwen-7B	COCO	+0.40	+0.00	-0.32	+0.43	+1.54
Qwen-7B	AOKVQA	+0.01	+0.40	-0.58	+0.03	+0.67

possible.” prompt) is *less* harmful than VCD and OLM in balanced accuracy on LLaVA GQA, confirming that the damage from YES inflation is not specific to any particular implementation.

5.5 The COCO Exception Confirms the Theory

COCO is the one case where YES-inflating methods improve balanced accuracy slightly. This is not discriminative improvement — the LLaVA baseline on COCO is YES-*deficient* (YES% $\approx$ 20%, far below the 25% ground truth on 25/75). The baseline’s operating point is already suboptimal; any YES inflation moves it toward the optimum, improving balanced accuracy until it crosses the optimal threshold.

The decisive evidence: PBA achieves +0.73% balanced accuracy at 25/75 and +0.72% at 75/25 on LLaVA COCO — nearly identical gains in both regimes. A prompt that says “Answer yes whenever possible.” produces the same profile. This is threshold correction, not hallucination reduction.

5.6 The ROC Curve Interpretation

Since POPE outputs are binary (yes/no), each method occupies a single point on the ROC plane. Our balanced accuracy IS the AUC for this binary classifier.

If CD methods lifted the ROC curve upward, balanced accuracy would increase vs. baseline consistently across all label ratios. Instead:

- On LLaVA GQA/AOKVQA: VCD and SID are below baseline in *both* regimes.
- The (TPR, TNR) operating point shifts — TPR rises, TNR falls — but the net balanced accuracy decreases, confirming the operating point moved *away* from the optimal balanced point.

- This is the definitive signature of a **threshold shift without curve lift**.

5.7 Why Qwen Is Different

For Qwen-7B, all balanced accuracy changes are within ± 1.4 points of baseline, regardless of method or regime (Table 14). This is consistent with a well-calibrated model already near its optimal operating point: any threshold shift, in either direction, produces only small balanced accuracy changes. The signal is too small to measure reliably.

This confirms: CD methods have no discriminative value independent of the base model’s YES bias. They are parasitic on LLaVA’s miscalibration, not on any genuine improvement in visual understanding.

5.8 Summary: The AUC Argument in One Paragraph

CD methods improve POPE accuracy by increasing YES response rates. Our balanced accuracy analysis proves this is a threshold-shift artifact, not discrimination improvement. When class imbalance is removed from the metric, VCD and SID on LLaVA GQA and AOKVQA perform *worse* than the no-intervention baseline — by up to 2.69 percentage points. Standard accuracy swings 7–68 $\times$ more than balanced accuracy across ratio regimes, meaning 86–98% of the reported accuracy variation is pure class-imbalance artifact. The ROC curve has not moved; the methods are sliding the operating point along the same curve, past the balanced optimum, in both the 25/75 and 75/25 directions. **For any real-world deployment where label balance is not guaranteed to be 50/50, these methods offer no reliable benefit and may actively degrade discrimination.**